

ALFAAL12332

## 1-Methylthio-2-propanone

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 1-Methylthio-2-propanone  
1-Methylthio-2-propanone

**Cat No. :** L12332  
**CAS No** 14109-72-9  
**Molecular Formula** C4 H8 OS

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
No information available

**Odor**  
Stench

**Emergency Overview**  
Flammable liquid and vapor. Air sensitive. Stench.

#### Classification of the substance or mixture

Flammable liquids.

Category 3

#### Label Elements



**Signal Word**

**Warning**

**Hazard Statements**  
H226 - Flammable liquid and vapor

## 1-Methylthio-2-propanone

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P242 - Use non-sparking tools

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P243 - Take action to prevent static discharges

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion.

**Health Hazards**

The product contains no substances which at their given concentration are considered to be hazardous to health.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its volatility. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces.

**Other Hazards**

Stench. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                    | CAS No     | Weight % |
|------------------------------|------------|----------|
| 2-Propanone, 1-(methylthio)- | 14109-72-9 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

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Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Storage**

Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS

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96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Exposure Controls

## Engineering Measures

None under normal use conditions. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. .

## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Neoprene       |                                   |                 |             |                       |
| Natural rubber |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

## Appearance

**Physical State** Liquid

**Odor** Stench

**Odor Threshold** No data available

**pH** No information available

**Melting Point/Range** No data available

**Softening Point** No data available

**Boiling Point/Range** 69 - 70 °C / 156.2 - 158 °F

**Flash Point** No information available

**Method -** No information available

**Evaporation Rate** No data available

**Flammability (solid,gas)** Not applicable

Liquid

**Explosion Limits** No data available

**SAFETY DATA SHEET****1-Methylthio-2-propanone**

|                                                |                          |                                        |
|------------------------------------------------|--------------------------|----------------------------------------|
| <b>Vapor Pressure</b>                          | No data available        |                                        |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                            |
| <b>Specific Gravity / Density</b>              | No data available        |                                        |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                 |
| <b>Water Solubility</b>                        | No information available |                                        |
| <b>Solubility in other solvents</b>            | No information available |                                        |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                        |
| <b>Autoignition Temperature</b>                | No data available        |                                        |
| <b>Decomposition Temperature</b>               | No data available        |                                        |
| <b>Viscosity</b>                               | No data available        |                                        |
| <b>Explosive Properties</b>                    |                          | explosive air/vapour mixtures possible |
| <b>Oxidizing Properties</b>                    | No information available |                                        |
| <b>Molecular Formula</b>                       | C4 H8 OS                 |                                        |
| <b>Molecular Weight</b>                        | 104.17                   |                                        |

**SECTION 10. STABILITY AND REACTIVITY**

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Stability</b>                | Air sensitive.                                                    |
| <b>Hazardous Reactions</b>      | None under normal processing.                                     |
| <b>Hazardous Polymerization</b> | No information available.                                         |
| <b>Conditions to Avoid</b>      | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>       | No information available.                                         |

**Hazardous Decomposition Products** None under normal use conditions.

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information**

|                                               |                                                           |
|-----------------------------------------------|-----------------------------------------------------------|
| <b>(a) acute toxicity;</b>                    |                                                           |
| <b>(b) skin corrosion/irritation;</b>         | No data available                                         |
| <b>(c) serious eye damage/irritation;</b>     | No data available                                         |
| <b>(d) respiratory or skin sensitization;</b> |                                                           |
| <b>Respiratory</b>                            | No data available                                         |
| <b>Skin</b>                                   | No data available                                         |
| <b>(e) germ cell mutagenicity;</b>            | No data available                                         |
| <b>(f) carcinogenicity;</b>                   | No data available                                         |
|                                               | There are no known carcinogenic chemicals in this product |
| <b>(g) reproductive toxicity;</b>             | No data available                                         |
| <b>(h) STOT-single exposure;</b>              | No data available                                         |

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|                                            |                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------|
| (i) STOT-repeated exposure;                | No data available                                                                   |
| Target Organs                              | No information available.                                                           |
| (j) aspiration hazard;                     | No data available                                                                   |
| Symptoms / effects, both acute and delayed | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |

## SECTION 12. ECOLOGICAL INFORMATION

|                                                                                              |                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects                                                                          | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.                                                                                   |
| Persistence and Degradability<br>Persistence                                                 | No information available<br>Persistence is unlikely, based on information available.                                                                                                                          |
| Bioaccumulative Potential                                                                    | Bioaccumulation is unlikely                                                                                                                                                                                   |
| Mobility in soil                                                                             | The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.                |
| Endocrine Disruptor Information<br>Persistent Organic Pollutant<br>Ozone Depletion Potential | This product does not contain any known or suspected endocrine disruptors<br>This product does not contain any known or suspected substance<br>This product does not contain any known or suspected substance |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                                                                                          |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                   |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition. |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.                                  |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                         |                            |
|-------------------------|----------------------------|
| UN-No                   | UN1224                     |
| Proper Shipping Name    | KETONES, LIQUID, N.O.S.    |
| Technical Shipping Name | (1-Methylthio-2-propanone) |
| Hazard Class            | 3                          |
| Packing Group           | III                        |

IMDG/IMO

|                         |                            |
|-------------------------|----------------------------|
| UN-No                   | UN1224                     |
| Proper Shipping Name    | KETONES, LIQUID, N.O.S.    |
| Technical Shipping Name | (1-Methylthio-2-propanone) |

## 1-Methylthio-2-propanone

**Hazard Class** 3  
**Packing Group** III

IATA

**UN-No** UN1224  
**Proper Shipping Name** KETONES, LIQUID, N.O.S.  
**Technical Shipping Name** (1-Methylthio-2-propanone)  
**Hazard Class** 3  
**Packing Group** III

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                    | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|--------|------|-----|-------|------|------|------|------|
| 2-Propanone, 1-(methylthio)- | -                                                   | -                                       | X    | -     | -      | -    | -   | -     | -    | -    | X    | -    |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 01-May-2024  
**Revision Summary** New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**1-Methylthio-2-propanone**

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**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**